

Dr.Saurabh J. Shigwan

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ASSISTANT PROFESSOR, COMPUTER SCIENCE AND ENGINEERING, SHIV NADAR IOE DELHI-NCR

PROFESSIONAL SUMMARY

Shiv Nadar Institute of Eminence Delhi-NCR
Designation: Assistant Professor, CSE department

Jan '21 - Present

Psychiatry Neuroimaging Laboratory, HMS, Boston
Designation: Pre-doctoral Fellow

Aug '19 - Mar '20

COMPUTER SKILLS

Languages: Python, MATLAB, Cython, C/C++
Platforms/Libraries: SciPy-NumPy, Pytorch, PyG, Tensorflow, Keras, DiPy
Research Tools: Slicer, ITK-SNAP, VTK

ONGOING RESEARCH PROJECTS

Open Set Recognition using Neural Collapse

Students: Arnav Aditya, Vishal Chaudhary

Sept '24 - Present

Collaborator: Dr. Nitin Kumar(SNIOE)

- Understanding Neural Collapse phenomenon
- Designing unique loss function to improve open world classification
- Implementation is in **Python-PyTorch**

Unsupervised Classification using Graph Neural Networks for Healthcare datasets

Students: Tejaswi Abburi

Sept '24 - Present

Collaborators: Dr. Nitin Kumar(SNIOE)

- Designed a SOTA method for unsupervised classification using GNN and Modularity loss
- Experimental results on ADNI and NIFD datasets.
- Compared result with SOTA methods.

Underwater Video Restoration and 3D Reconstruction for Marine Ecological Conservation

Student: Diksha Dhillon

Apr '26 - Present

Collaborators: Dr. Sumit Shekhar (SNIOE)

- To develop data processing methodologies that improve the quality of underwater videos
- Developing a physics-guided and learning-based restoration framework to correct color distortions and enhance underwater video clarity.
- Creating a benchmark dataset linking raw, restored, and reconstructed data with environmental metadata

RESEARCH INTERESTS

Statistical Modeling and Inference, Medical Image Processing, Bayesian Analysis, Machine Learning, Computer Vision, Deep Learning, Convolution network, Graph convolutional network, Shape analysis.

AWARDS & ACHIEVEMENTS

- 1) Secured **Research Funding of \$16000** from **Mass General Brigham** to do research at Harvard Medical lab on **Brain Tractography using Diffusion-MRI**.
- 2) Dr. Abhishek Tiwari has been graduated under my co-supervision along with Prof. Rajeev Kuamr in 2024.

PUBLICATIONS

VSS Tejaswi Abburi, Ananya Singhal, **Saurabh J. Shigwan** and Nitin Kumar, "ARMARecon: An ARMA Convolutional Filter based Graph Neural Network for Neurodegenerative Dementias Classification", 23rd IEEE International Symposium on Biomedical Imaging (ISBI), 2026

Arnav Aditya, Nitin Kumar, and **Saurabh J. Shigwan**. "UCDSC: Open Set UnCertainty aware Deep Simplex Classifier for Medical Image Datasets." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision(WACV), 2026.

Dubey, Shikhar, Sourava Kumar Behera, Krish Mittal, Manikandan Ravikiran, Nitin Kumar, **Saurabh J. Shigwan**, and Rohit Saluja, "Multi-Feature Graph Convolution Network for Hindi OCR Verification." 1st Workshop on Benchmarks, Harmonization, Annotation, and Standardization for Human-Centric AI in Indian Languages (BHASHA 2025). 2025.

A. Mudit Adityaja, **Saurabh J. Shigwan**, and Nitin Kumar, "UnSegMedGAT: Unsupervised Medical Image Segmentation using Graph Attention Networks Clustering", 22nd IEEE International Symposium on Biomedical Imaging (ISBI), 2025

Kovvuri Sai Gopal Reddy, Bodduluri Saran, A. Mudit Adityaja, **Saurabh J. Shigwan**, Nitin Kumar, and Snehasis Mukharjee, "UnSeGArmaNet: Unsupervised Image Segmentation using Graph Neural Networks with Convolutional ARMA Filters", 35th British Machine Vision Conference (BMVC), 2024

Abhishek Tiwari, Rajeev Kumar Singh and **Saurabh J. Shigwan**, "SwinDTI: swin transformer-based generalized fast estimation of diffusion tensor parameters from sparse data" Neural Computing and Applications, Springer, 2023

Abhishek Tiwari, Ananya Singhal, **Saurabh J. Shigwan**, Rajeev Kumar Singh, "Early Diagnosis of Alzheimer through Swin-Transformer-Based Deep Learning Framework using Sparse Diffusion Measures" The 15th Asian Conference on Machine Learning (ACML 2023)

Abhishek Tiwari, Ananya Singhal, **Saurabh J. Shigwan**, Rajeev Kumar Singh, "Deep Learning Framework using Sparse Diffusion MRI for Diagnosis of Frontotemporal Dementia", IEEE/CVF International Conference on Computer Vision, BioImage Computing Workshop, ICCV 2023

Abhishek Tiwari, **Saurabh J. Shigwan** and Rajeev Kumar Singh, et.al., "Validation of Deep Learning techniques for quality augmentation in diffusion MRI for clinical studies" Elsevier NeuroImage: Clinical Q1 SCI Journal Impact Factor = 4.2

Saurabh J. Shigwan, Akshya Gaikwad, Suyash P. Awate, "Object Segmentation With Deep Neural Nets Coupled with a Shape Prior, When Learning from a Training Set of Limited Quality and Small Size" to appear in *International Symposium on Biomedical Imaging (ISBI-2020)*, Iowa City, USA

Saurabh J. Shigwan, Suyash P. Awate, "Hierarchical generative modeling and Monte-Carlo EM in Riemannian shape space for hypothesis testing" appeared in *Medical Image Computing and Computer Assisted Intervention (MICCAI-2016)*, Athens, Greece

Akshya Gaikwad, **Saurabh J. Shigwan**, Suyash P. Awate, "A statistical model for smooth shapes in Kendall shape space" appeared in *Medical Image Computing and Computer Assisted Intervention (MICCAI-2015)*, Munich, Germany

EDUCATION

PhD, Computer Science (CGPA: 8.85/10)

Jul' 14 - August' 20

CSE, Indian Institute of Technology Bombay, Maharashtra, India

Thesis title: **Hierarchical Pointset-Based Statistical Shape Modeling and Applications**

MTech, Computer Science (I Class)
MIU, Indian Statistical Institute Kolkata, West Bengal, India

Jul' 12 - Jul' 14

BE, Computer Engineering (I Class)
University of Mumbai, Maharashtra, India

Jul' 07 - Jul' 11